

- 2 The Labrador retriever is a breed of dog that, as an adult, shows continuous variation in body mass.

Figure 2.1 shows a Labrador retriever.



Figure 2.1

- (a) (i) Explain what is meant by continuous variation.

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- (ii) Describe the causes of continuous variation in the body mass of adult mammals, such as Labrador retriever dogs.

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- (b) Scientists have discovered that the gene *POMC* helps to regulate the body mass of Labrador retrievers.

Approximately 25% of Labrador retrievers have at least one mutant *POMC* allele.

Figure 2.2 shows the mean body mass of Labrador retrievers with each of the three possible genotypes for this gene:

- two normal *POMC* alleles (homozygous normal)
- one normal *POMC* allele and one mutant *POMC* allele (heterozygous)
- two mutant *POMC* alleles (homozygous mutant).

Each error bar shows the mean ± 1 standard deviation.

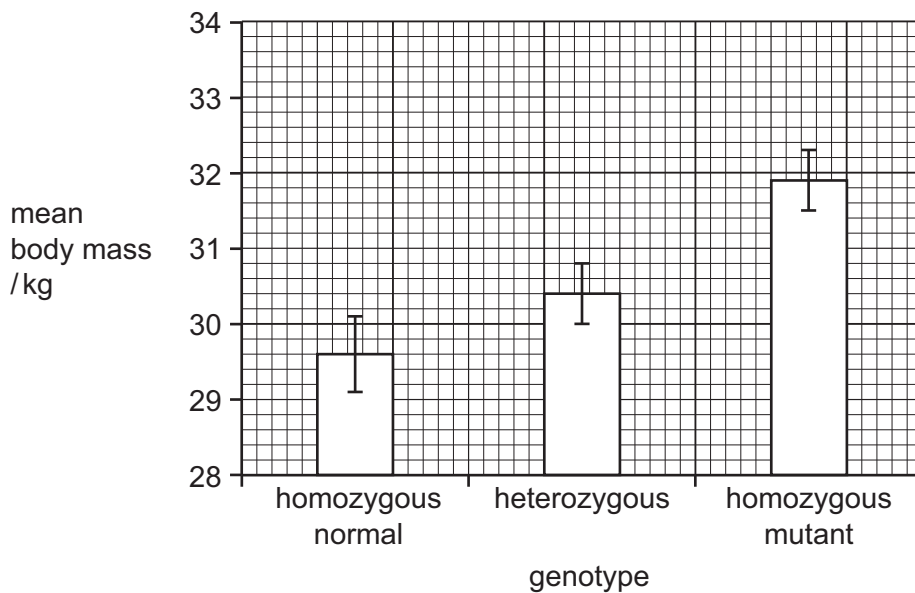


Figure 2.2

- (i) With reference to Figure 2.2, describe the effect of the mutant *POMC* allele on the body mass of Labrador retrievers.

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- (ii) Labrador retrievers with mutant *POMC* alleles have changes in behaviour that affect body mass.

Suggest a change in behaviour in Labrador retrievers with mutant *POMC* alleles that would affect body mass.

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[Total: 7]